DBMS Transactions & ACID Properties

1. What is a Transaction?

Definition

A transaction is a sequence of database operations performed as a single logical unit of work.

A transaction may contain:

- Read operation
- Write operation
- Update operation
- Delete operation

Either:

- All operations execute successfully
  OR
- None of them execute

Real-Life Example

Bank Transfer:

A → B : ₹1000

Steps:

1. Deduct ₹1000 from A
2. Add ₹1000 to B

If system crashes after deduction but before addition:
Money lost.

Transaction ensures both operations happen together.

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2. Transaction States

A transaction passes through multiple states.

A. Active State

Transaction is executing.

Example:
SQL queries running.

B. Partially Committed State

Last statement executed but changes not permanently saved yet.

C. Committed State

Changes permanently stored in database.

Transaction successful.

D. Failed State

Transaction cannot proceed because of:

- System crash
- Power failure
- Error

E. Aborted State

All changes rolled back.

Database restored to previous state.

Transaction State Diagram

Active → Partially Committed → Committed
  ↓
 Failed → Aborted

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3. ACID Properties

ACID properties ensure reliable transactions.

A = Atomicity

Definition

Either all operations execute or none execute.

"ALL OR NOTHING"

Example

Bank transfer:

Deduct ₹1000 from A
Add ₹1000 to B

If second step fails:
First step must also undo.

Atomicity Problem Example

Without atomicity:

Account	**Balance**
A	5000
B	2000

After deducting from A:

| A | 4000 |
| B | 2000 |

System crashes.

Money lost permanently.

Solution

Rollback restores:

| A | 5000 |
| B | 2000 |

B = Consistency

Definition

Database must remain in a valid state before and after transaction.

Example

Total bank money:

Before = ₹7000
After = ₹7000

Database rules must remain satisfied.

C = Isolation

Definition

Multiple transactions should not interfere with each other.

Concurrent transactions behave as if executed one by one.

Example

Two users booking same seat simultaneously.

Without isolation:
Both may get same seat.

Isolation prevents this.

D = Durability

Definition

Once transaction commits, data remains permanently saved even after crash.

Example

Money transfer successful.

Even if power fails:
Changes remain stored.

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4. Transaction Operations

A. Read Operation

Syntax

READ(X)

Reads data item X.

B. Write Operation

Syntax

WRITE(X)

Writes updated value.

Example

READ(A)
A = A - 1000
WRITE(A)

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5. Schedule in DBMS

Definition

Order in which operations of multiple transactions execute.

Types of Schedules

A. Serial Schedule

Transactions execute one after another.

Example

T1 → T2

No overlap.

Safe but slow.

B. Concurrent Schedule

Transactions execute simultaneously.

Example

T1 and T2 overlap

Fast but may cause problems.

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6. Problems in Concurrent Transactions

A. Lost Update Problem

Two transactions update same data.

One update overwritten.

Example

Initial balance = 1000

T1 adds 100
T2 subtracts 50

Final balance may become incorrect.

B. Dirty Read

Transaction reads uncommitted data.

Example

T1 updates balance but not committed.
T2 reads updated value.
T1 later rolls back.

T2 used invalid data.

C. Non-Repeatable Read

Same query gives different results inside same transaction.

Example

T1 reads salary = 5000
T2 updates salary = 7000
T1 reads again → 7000

Different values obtained.

D. Phantom Read

Rows appear/disappear during transaction.

Example

T1 counts employees = 10
T2 inserts new employee
T1 counts again = 11

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7. Concurrency Control

Definition

Techniques used to manage simultaneous transactions safely.

Goal:

- Maintain consistency
- Prevent conflicts

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8. Lock-Based Protocol

Transactions lock data before access.

Types of Locks

A. Shared Lock (S-Lock)

Definition

Used for reading.

Multiple transactions can read simultaneously.

Example

READ only

B. Exclusive Lock (X-Lock)

Definition

Used for writing.

No other transaction can read/write.

Example

WRITE operation

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9. Two-Phase Locking (2PL)

Definition

Transaction has two phases:

A. Growing Phase

Locks acquired.

No locks released.

B. Shrinking Phase

Locks released.

No new locks acquired.

Example

Lock(A)
Lock(B)
Unlock(A)
Unlock(B)

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10. Deadlock

Definition

Two transactions wait for each other forever.

Example

T1 holds A, waits for B
T2 holds B, waits for A

Both stuck.

Deadlock Prevention

- Timeout
- Resource ordering
- Wait-Die scheme
- Wound-Wait scheme

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11. Timestamp Protocol

Definition

Each transaction gets unique timestamp.

Older transaction gets priority.

Ensures serializability.

Example

T1 = older
T2 = newer

If conflict occurs:
Older transaction executes first.

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12. Serializability

Definition

Concurrent schedule should produce same result as serial schedule.

Types

A. Conflict Serializability

Based on conflicting operations.

B. View Serializability

Based on final database view.

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13. Recoverability

Definition

Ability to recover database after failure.

Types

A. Recoverable Schedule

Commit occurs safely.

B. Cascading Rollback

Failure of one transaction causes rollback of others.

C. Cascadeless Schedule

Avoids cascading rollback.

Best approach.

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14. Checkpointing

Definition

Saving database state periodically.

Used for recovery.

Example

After every 100 transactions:
System saves checkpoint.

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15. Log-Based Recovery

Definition

System maintains log file of all operations.

Log Example

<T1, A, 5000, 4000>

Meaning:
A changed from 5000 to 4000.

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16. Commit and Rollback

COMMIT

Permanently saves changes.

COMMIT;

ROLLBACK

Undo changes.

ROLLBACK;

17. Important SQL Transaction Commands

BEGIN TRANSACTION

Starts transaction.

BEGIN;

COMMIT

COMMIT;

ROLLBACK

ROLLBACK;

SAVEPOINT

Creates temporary rollback point.

SAVEPOINT sp1;

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18. Most Important Interview Questions

Q1. What is transaction?

Sequence of operations treated as single logical unit.

Q2. What are ACID properties?

- Atomicity
- Consistency
- Isolation
- Durability

Q3. Difference between shared lock and exclusive lock?

Shared Lock	Exclusive Lock
Read only	Read + Write
Multiple allowed	Only one allowed

Q4. What is deadlock?

Transactions waiting for each other forever.

Q5. What is serializability?

Concurrent execution equivalent to serial execution.

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19. Shortcut Tricks for Exams

Atomicity

All or nothing.

Consistency

Valid database state.

Isolation

Transactions independent.

Durability

Changes permanent.

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20. Placement Focus Topics

Most important:

- ACID properties
- Deadlock
- Locks
- Serializability
- Recoverability
- Concurrency problems
- Commit & Rollback
- 2PL Protocol